Computer Organization and Architecture
(Code: CS256)

Lecture-30: Instruction Set Architecture

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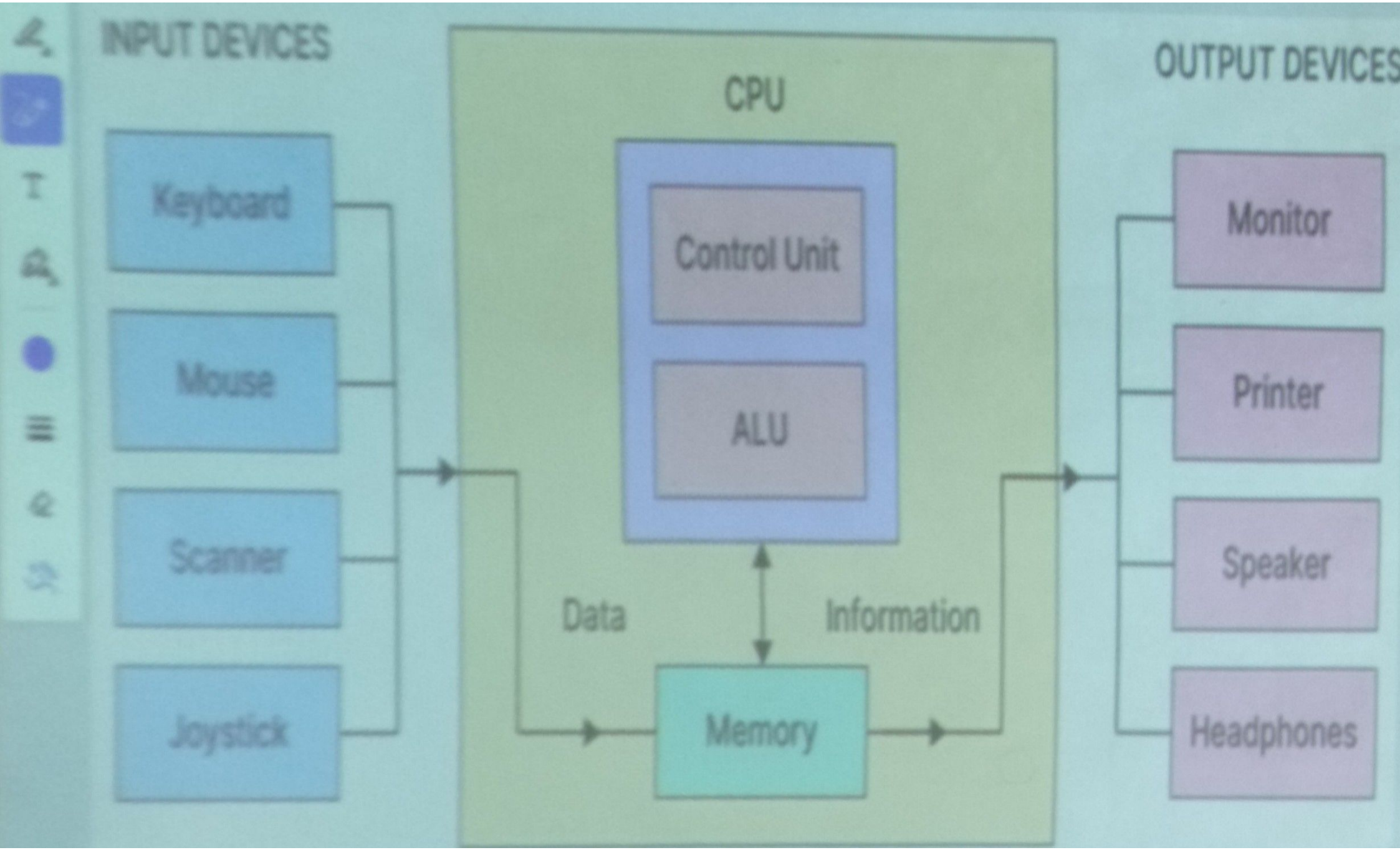
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CPU
Control Unit
ALU
Data
Information
Memory
OUTPUT DEVICES
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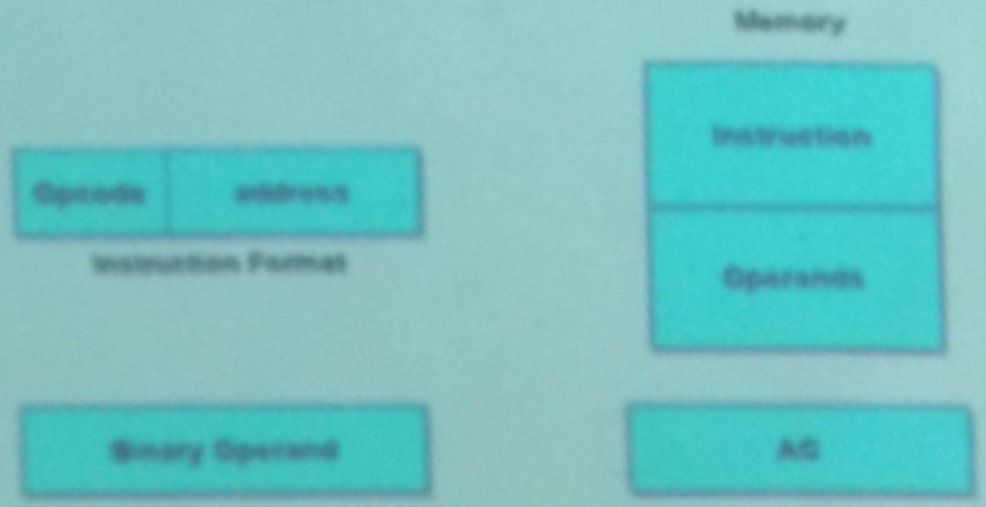
Printer
Speaker
Headphones

Computer Architecture

- *Computer architecture* refers to those attributes of a system visible to a programmer or, those attributes that have a direct impact on the logical execution of a program.
- *A term that is often used interchangeably with computer architecture is instruction set architecture (ISA).*
- The ISA defines instruction formats, instruction opcodes, registers, instruction and data memory; the effect of executed instructions on the registers and memory; and an algorithm for controlling instruction execution.



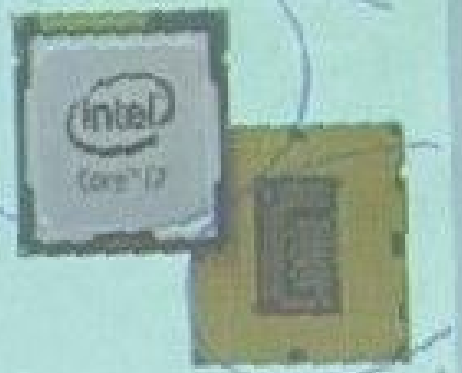
Computer Architecture

- Examples of architectural attributes include the instruction set, the number of bits used to represent various data types (e.g., numbers, characters), I/O mechanisms, and techniques for addressing memory.
- Examples
  - All Intel x86 family share the same basic architecture
  - The IBM System/370 family share the same basic architecture

Symbol	Hex Code I = 0	Hex Code I = 1	Description
AND	0xxx	8xxx	AND memory word to AC
ADD	1xxx	9xxx	Add memory word to AC
LDA	2xxx	Axxx	Load AC from memory
STA	3xxx	Bxxx	Store content of AC into memory
BUN	4xxx	Cxxx	Branch unconditionally
BSA	5xxx	Dxxx	Branch and save return address
ISZ	6xxx	Exxx	Increment and skip if zero
CLA	7800		Clear AC
CLE	7400		Clear E
CMA	7200		Complement AC
CME	7100		Complement E
CIR	7080		Circulate right AC and E
CIL	7040		Circulate left AC and E
INC	7020		Increment AC
SPA	7010		Skip next instr. if AC is positive
SNA	7008		Skip next instr. if AC is negative
SZA	7004		Skip next instr. if AC is zero
SZE	7002		Skip next instr. if E is zero
HLT	7001		Halt computer
INP	F800		Input character to AC
OUT	F400		Output character from AC
SKI	F200		Skip on input flag
SKO	F100		Skip on output flag
ION	F080		Interrupt on
IOF	F040		Interrupt off

Microprocessor

- A microprocessor is an IC that executes arithmetic and logical functions that are defined by a program inside the system.
- Also, *a computer system that is employed with a microprocessor is termed as a microcomputer.* It is basically an LSI i.e., large scale integration chip.
- Each and everything in a processor is stored in the form of 0's and 1's. As it is designed to operate in binary digits i.e., bits.
- The instructions that are performed by a microprocessor is in the form of bits, hence, is not in human understandable format.
- This machine level language is only understood by the microprocessor inside a system.
- Basically, it is classified according to its word length. So, we can say if the processor forms an 8-bit word then it is an 8-bit microprocessor. Similarly, a processor with the 32-bit word is a 32-bit microprocessor.
- *A microprocessor resembles a CPU but a microprocessor consists of logic circuitry with the control unit on a single chip.*



Microprocessor

- The microprocessor was introduced in 1971 by Intel Corporation, called Intel 4004.
- Basically, that was the first time when a processor was manufactured on a single chip. That processor was capable of performing arithmetic and logical operations like addition, subtraction etc.
- Further, with the coming time, an improved version of the 4-bit microprocessor was again introduced by Intel, called 4040, followed by many other 4 bits microprocessor.
- In 1973, intel again invented a microprocessor that can be operated on the 8-bit word, this was called 8008. But further, its advancement was introduced by Intel, called 8080.
- But, this was not the end as 32-bit and 64-bit microprocessors were also developed.

Computer Architects

- ISA refers to the actual programmer visible machine interface
- One can think of a ISA as a hardware functionality of a given computer.
- Computer architects must design a computer to meet functional requirements as well as price, power, and performance goals.
- Often, they also have to determine what the functional requirements are, which can be a major task.
- Once a set of functional requirements has been established, the architect must try to optimize the design.

Specifying a Program and Its Instructions

- High-level programs: each statement translates to many instructions
  - E.g., c ← a + b to ===>

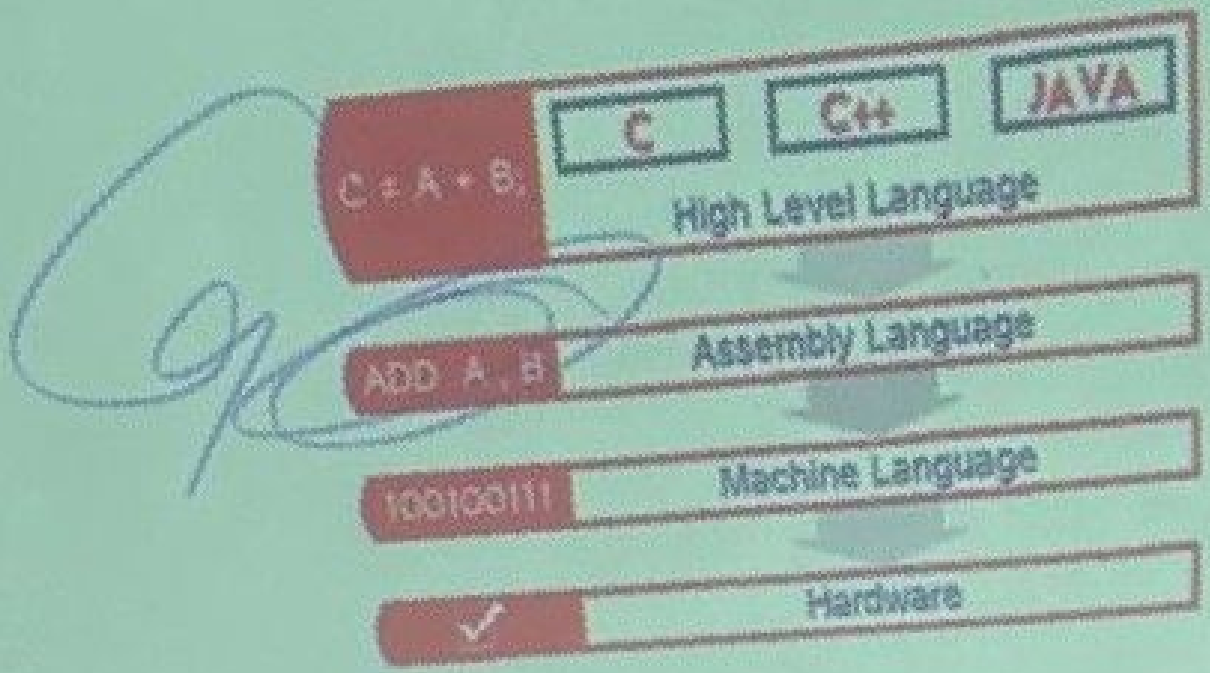
Load a into r1
Load b into r3
r2 ← r1 + r3
Store r2 into c

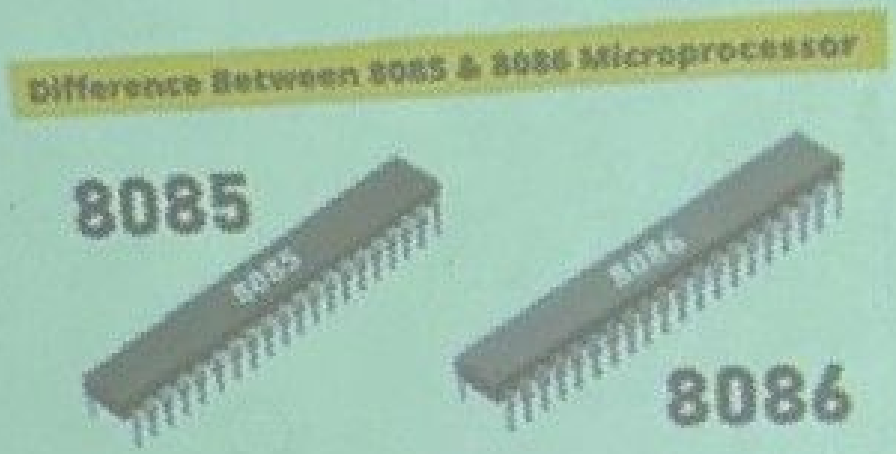
- Assembly language: specify each machine instruction, using mnemonic form
  - E.g. Load r1, A
- Machine language: specify each machine instruction, using bit patterns
  - E.g. 11011010000011110011

Machine/Assembly Language

We have a machine that can execute instructions

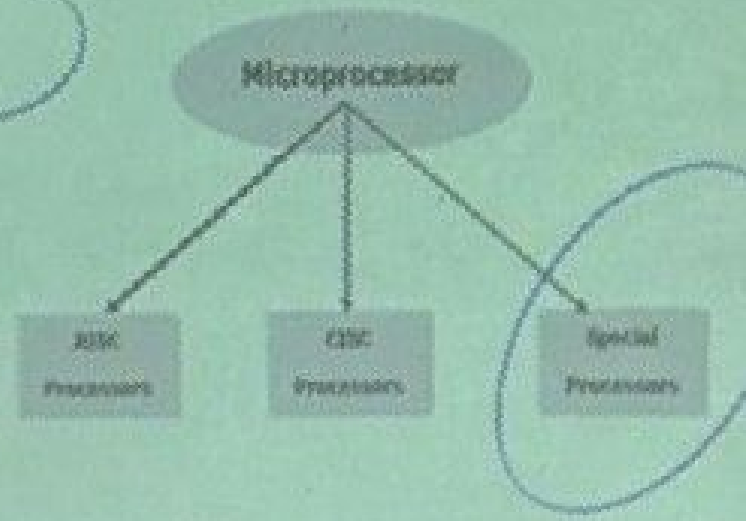
- Basic Questions:
  - What instructions?
  - How are these instructions represented to the computer hardware?
    - Simple or complex instructions?

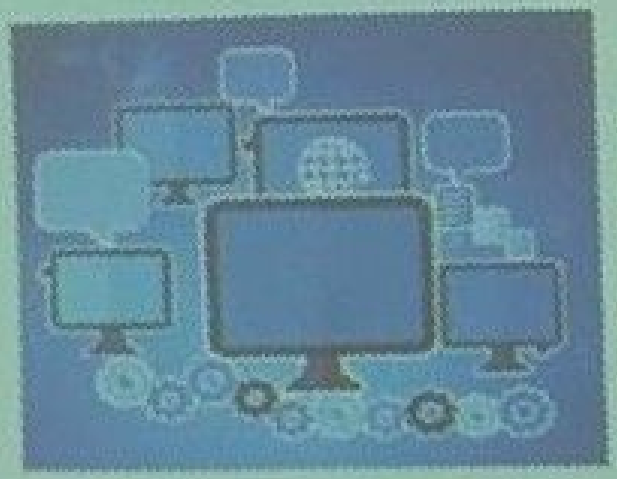


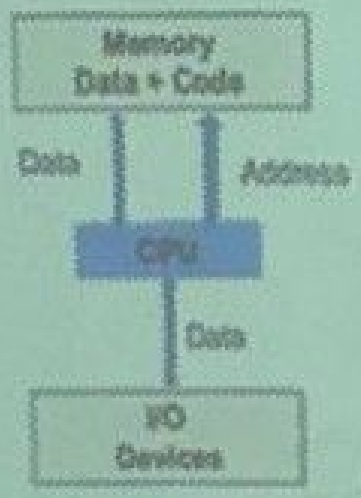


Types of Computer Architecture

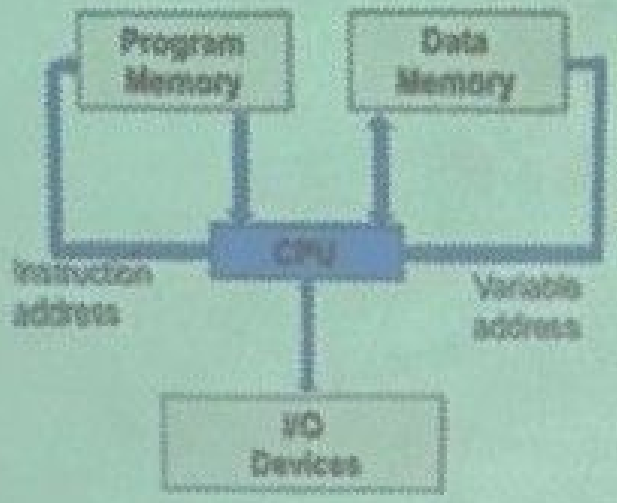
- **Instruction Set Characteristics**
  - Complex Instruction Set Computer (CISC)
  - Reduced Instruction Set Computer (RISC)
- **Memory Design** (Split and Unified)
  - Von Neumann Architecture
  - Harvard Architecture





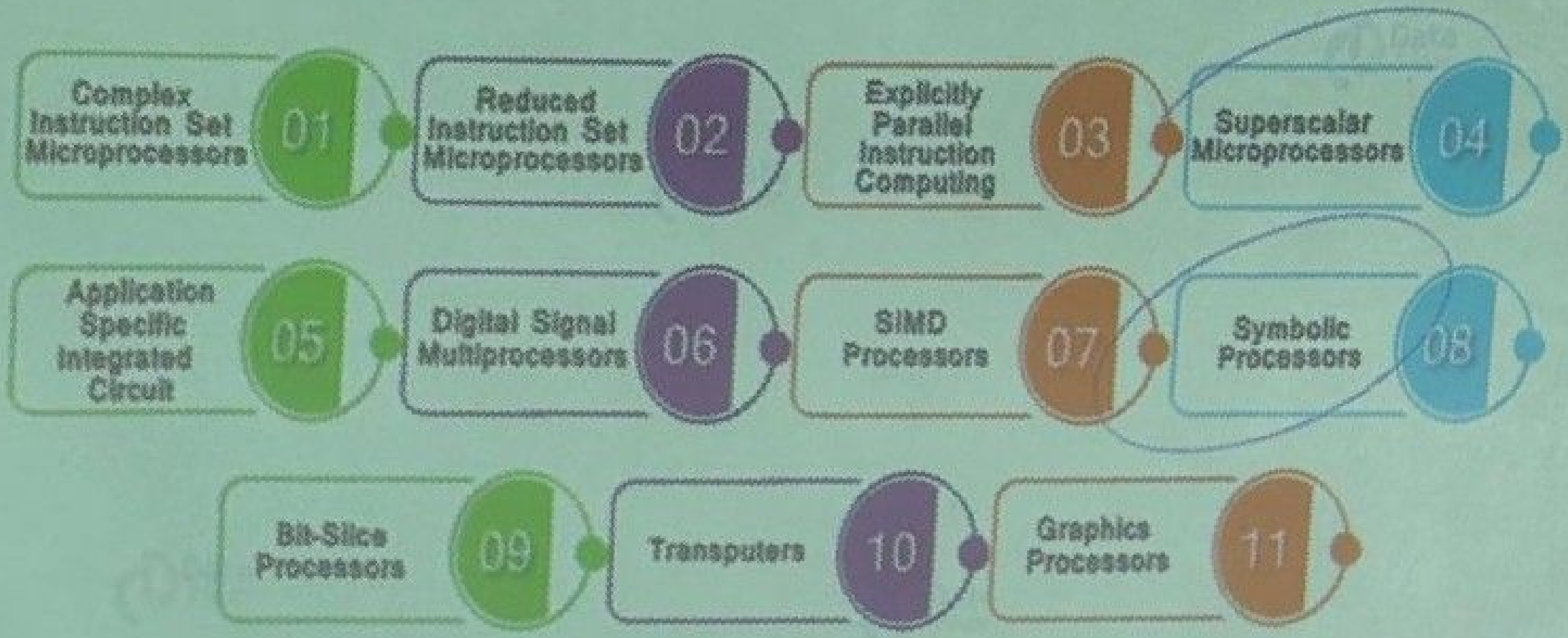


Von Neumann Machine



Harvard Machine

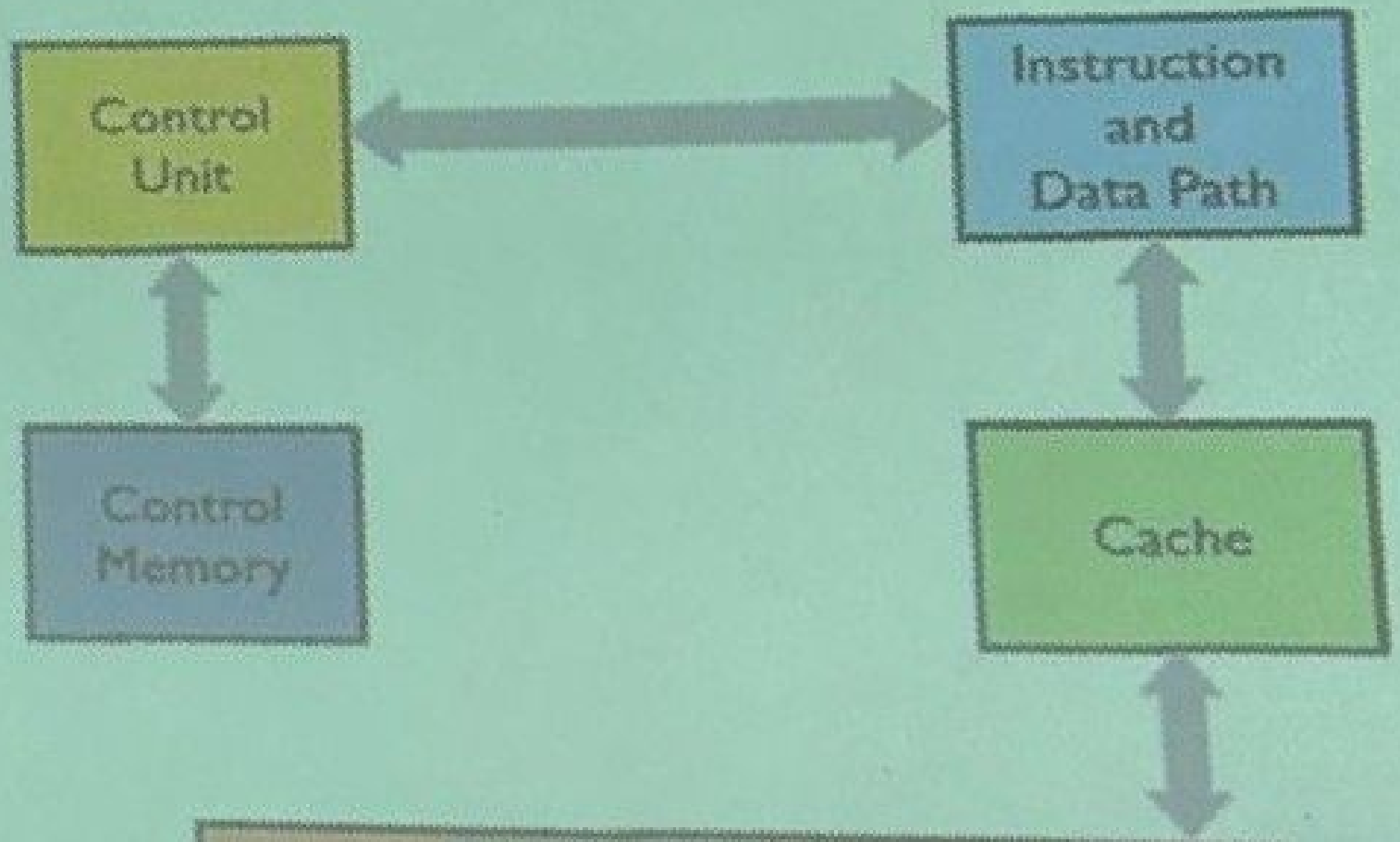
Types of Microprocessor



Complex Instruction Set Computer

- *CISC was introduced in 1970 and it handles instruction execution without making use of multiple codes.*
- It is based on the complex nature of instruction set architecture.
- A complex instruction set computer (CISC) is a computer architecture in which single instructions can execute several low-level operations such as
  - a load from memory,
  - an arithmetic operation, and
  - a memory store
- *Machine is capable of multi-step operations or addressing modes within single instructions.*

Complex Instruction Set Computer



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CISC Architecture

CISC Instructions

```
a[i++] = a[i] + b[i];
b[i]  = b[--i] - 1;
```

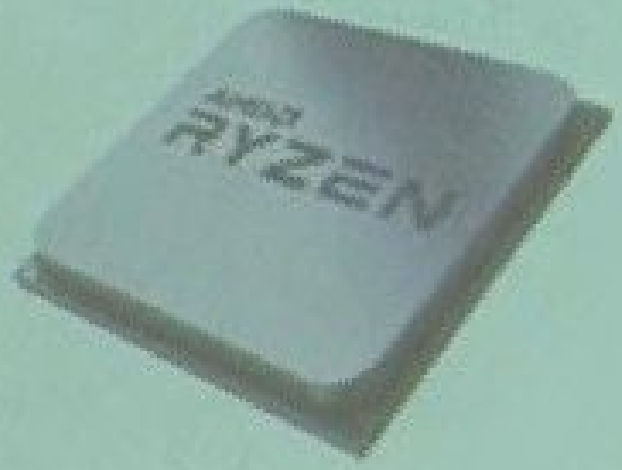
CISC Code:

```
add   (R3)+, (R3), (R4)
sub (R4),  -(R4),  1
```

CISC Processors

- Examples: Most common microprocessor designs such as the Intel 80x86, Motorola 68K, VAX series followed the CISC philosophy.
- CISC processors are mainly associated with real-time applications such as security systems, home automation, etc.

- Modern Intel i5, i7, … are CISC Processors
- AMD Ryzen Series are CISC Processors.



Features of CISC Processor

- **Decoding**: The instructions are of complex nature, thus offers complex instruction decoding.
- **Instruction size**: The instructions within this processor are large in size and are of variable nature.
- **Data Types and Addressing Modes**: It involves more data types and complex addressing modes.
- **General Purpose Registers**: The operations are performed in the memory itself and the results of operations are also stored within the memory so due to this reason less general-purpose registers are needed in the case of CISC processors.
- **Clock Cycle**: As the size of instructions here is variable thus more than one clock cycle may be used for instruction execution.

CISC Performance

- Consider the following CISC program segment.
- Assume
  - Each **mov** instruction takes 1 cycle.
  - Each **mul** instruction takes 30 cycles.

10 RS

```
mov ax, 10
mov bx, 5
mul bx, ax
```

The total clock cycles for the CISC version might be:

(2 movs × 1 cycle) + (1 mul × 30 cycles) = 32 cycles

Adv. And Disadv. of CISC

- **Advantages**
  - The code sizes are small in the case of the CISC processors.
  - Compiler's work is reduced to a large extent high-level language statement is not required to be converted into assembly language.
  - The shortcode length supports less RAM requirement.

- **Disadvantages**
  - The high complexity makes the system somewhat inefficient.
  - The instruction is executed in more than one clock cycle.
  - *It does not support instruction pipelining.*

RISC

CISC → RISC

- Historically, machines tend to add features over time instruction opcodes
- Examples: IBM 70X, 70X0 series went from 24 opcodes to 185 in 10 years same time performance increased 30 times
  - addressing modes
  - special purpose registers
- Motivations are to improve efficiency, since complex instructions can be implemented in hardware and execute faster make life easier for compiler writers support more complex higher-level languages.
- Examination of actual code indicated many of these features were not used.

CISC → RISC

- Advocates to propose
  - simple, limited instruction set
  - large number of general purpose registers
  - and mostly register operations
  - optimized instruction pipeline
- Benefits should include
  - faster execution of instructions commonly used
  - faster design and implementation

1984

Motorola 68000
(CISC)

1994

Motorola PowerPC
(RISC)

2005

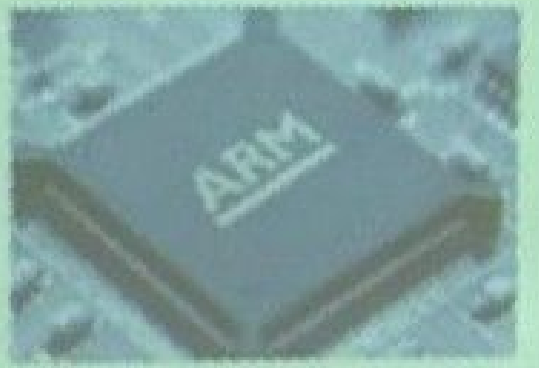
Intel x86
(CISC)

2020

Apple Silicon (ARM)
(RISC)

Reduced Instruction Set Computer

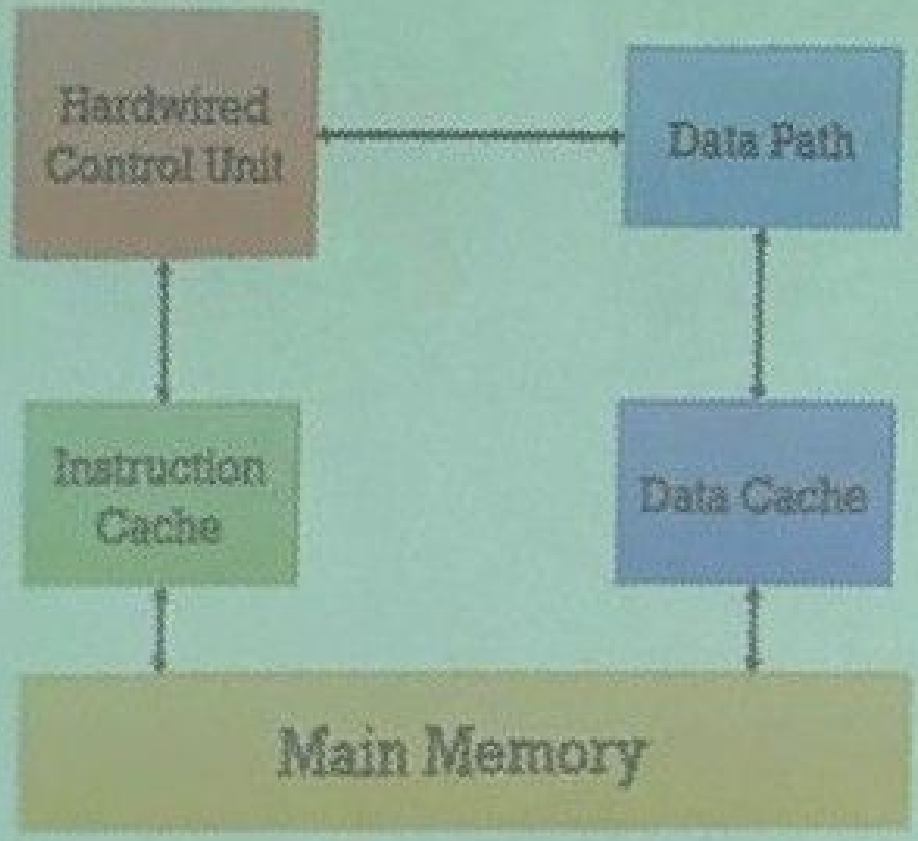
- After 1985, most computers announced have been of RISC architecture.
- RISC designers focused on two critical performance techniques in computer design:
  - the exploitation of instruction-level parallelism, first through pipelining and later through multiple instruction issue,
  - the use of cache, first in simple forms and later using sophisticated organizations and optimizations.



Reduced Instruction Set Computer

- *RISC, or Reduced Instruction Set Computer, is a type of microprocessor architecture that utilizes a small, highly-optimized set of instructions, rather than a more specialized set of instructions often found in other types of architecture.*
- Certain design features have been characteristic of most RISC processors:
  - **one cycle execution time:** RISC processors have a CPI (clock per instruction) of one cycle. This is due to the optimization of each instruction on the CPU and a technique called PIPELINING
  - **pipelining:** a technique that allows for simultaneous execution of parts, or stages, of instructions to more efficiently process instructions;
  - **large number of registers:** the RISC design philosophy generally incorporates a larger number of registers to prevent in large amounts of interactions with memory

RISC Architecture



RISC Architecture

RISC Processors

- Arm Architectures – ARM7, ARM9
- RISC-V
- Intel - Intel i860, Intel i960
- Motorola 88000
- The Sun Microsystems UltraSPARC.
- RISC processors are used in supercomputers, such as the Fugaku.

RISC Instruction Example

$A[i+1] = A[i] + M[i]$

$M[i] = M[i-1] - 1;$

RISC Code:

```
lw R1, 0(R3)
lw R2, 0(R4)
add R5, R1, R2
sub R2, R2, 1
sw 0(R3), R5
sw 0(R4), R2
```

RISC Instruction Example 2

Program to evaluate $X = (A + B) * (C + D)$

LOAD	R1, A	$R1 \leftarrow M[A]$
LOAD	R2, B	$R2 \leftarrow M[B]$
LOAD	R3, C	$R3 \leftarrow M[C]$
LOAD	R4, D	$R4 \leftarrow M[D]$
ADD	R1, R1, R2	$R1 \leftarrow R1 + R2$
ADD	R3, R3, R4	$R3 \leftarrow R3 + R4$
MUL	R1, R1, R3	$R1 \leftarrow R1 * R3$
STORE	X, R1	$M[X] \leftarrow R1$

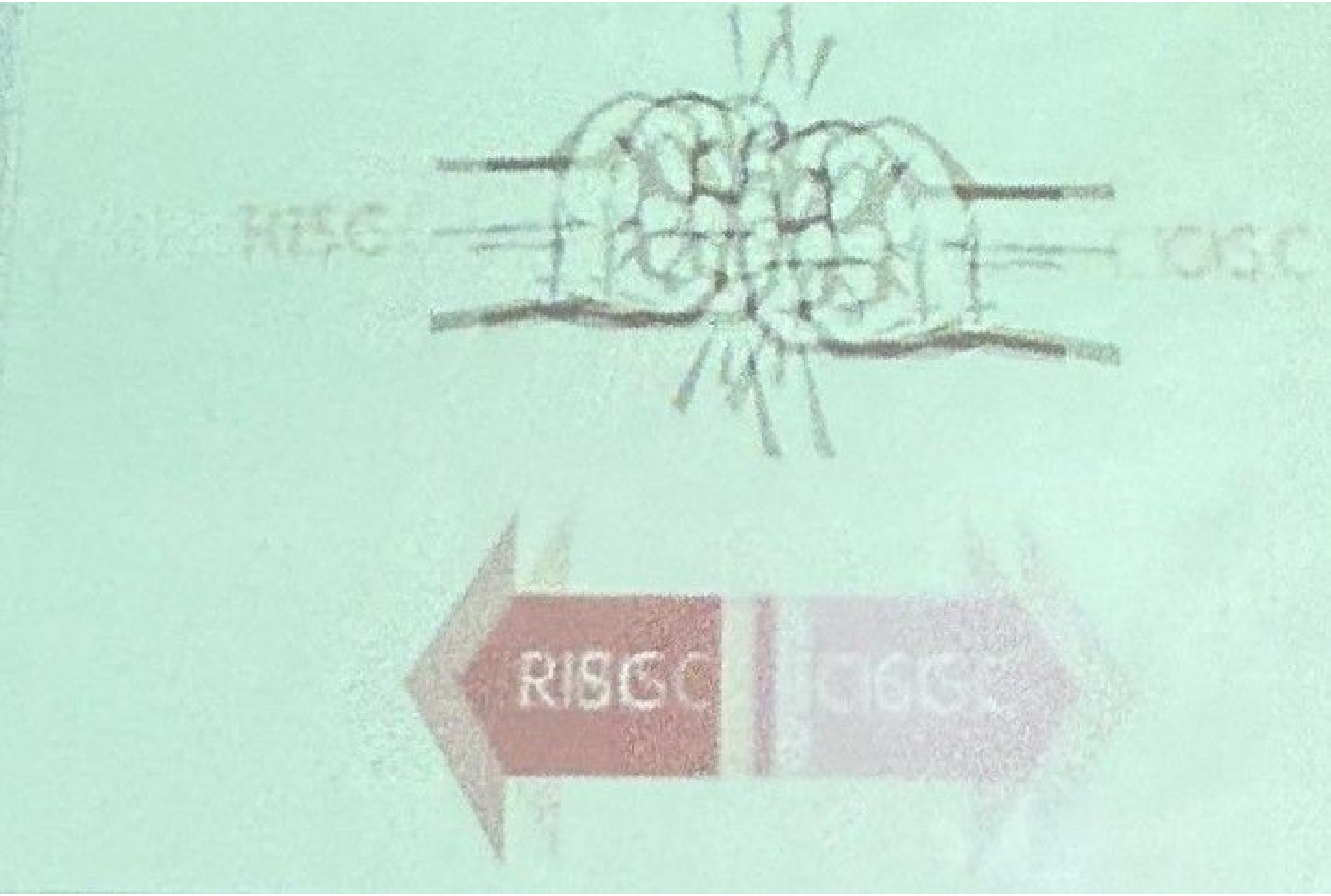
RISC Performance

- Consider the following RISC program segment.
- Assume
  - Each instruction takes 1 cycle.

```
        mov ax, 0
        mov bx, 10
        mov cx, 5
Begin   add ax, bx
        loop Begin
```

While the clock cycles for the RISC version is:

(3 moves × 1 cycle) + (5 adds × 1 cycle) + (5 loops × 1 cycle) = 13 cycles


RISC
CISC

RISC vs. CISC

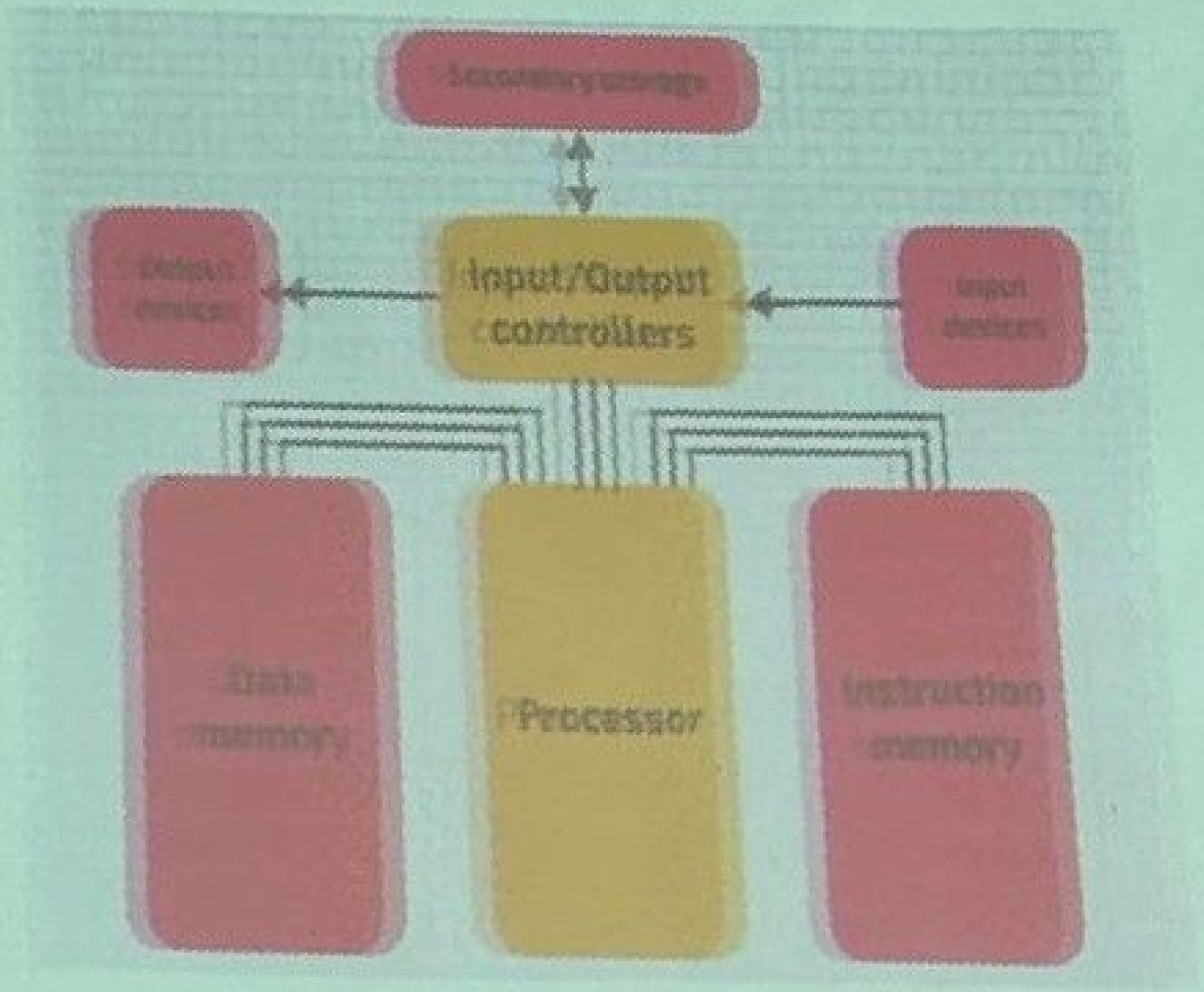
- It is a microprocessor architecture that uses small instruction set of uniform length.
- These simple instructions are executed in one clock cycle.
- These chips are relatively simple to design.
- They are inexpensive.
- Examples of RISC chips include SPARC, POWER PC.
- It has less number of instructions.
- It has fixed-length encodings for instructions.
- Simple addressing formats are supported.
- Registers are used for procedure arguments and return addresses.

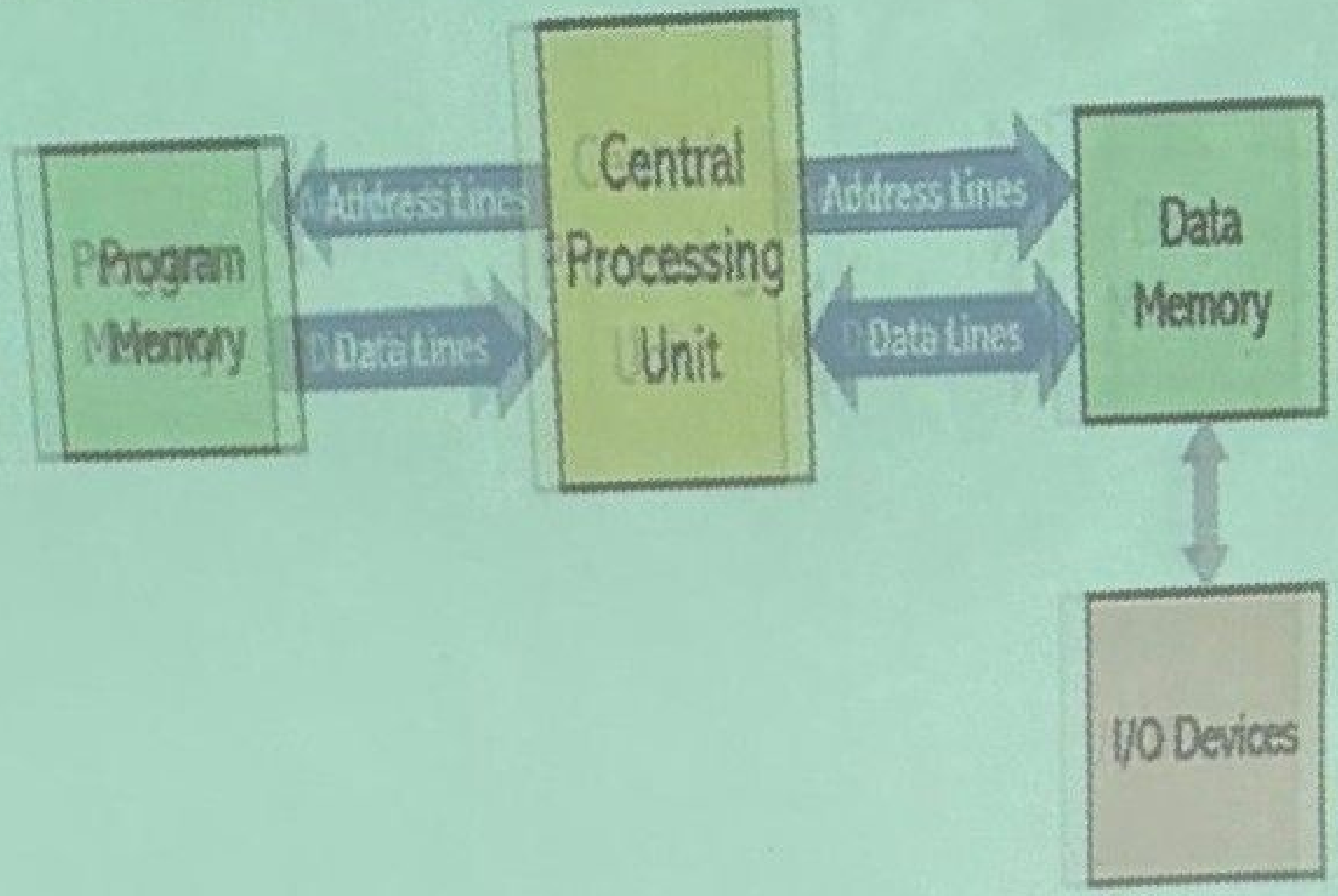
- This offers hundreds of instructions of different sizes to the users.
- This architecture has a set of special purpose circuits which help execute the instructions at a high speed.
- These chips are complex to design.
- They are relatively expensive.
- Examples of CISC include Intel architecture, AMD.
- It has more number of instructions.
- It has variable-length encodings of instructions.
- The instructions interact with memory using complex addressing modes.
- The stack is used for procedure arguments and return addresses.

Harvard Architecture

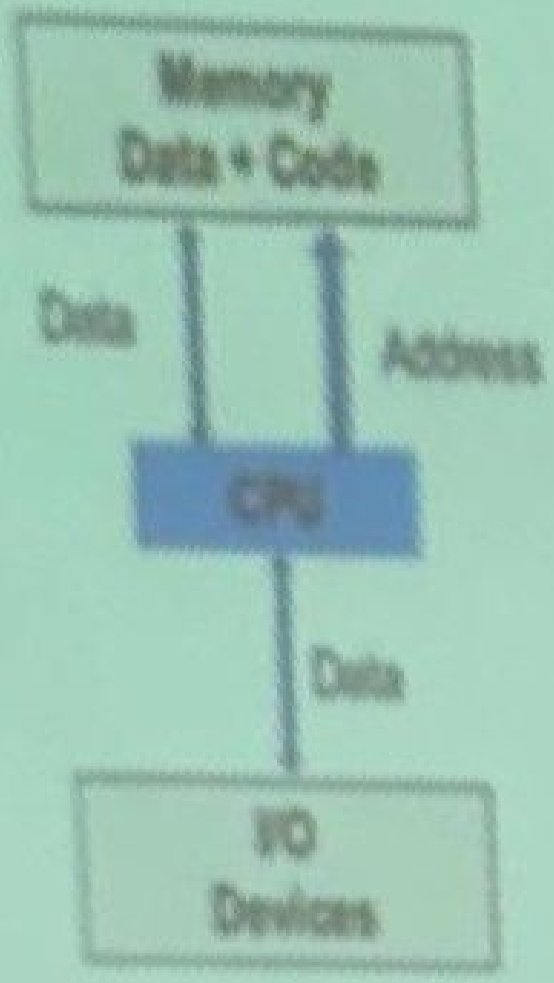
- Invented at Harvard for the Harvard Mark I computer in 1944.
- Harvard architecture keeps instructions and data in separate memories.
- The processor accesses these memories using separate data and address buses; the processor is connected to the 'instruction memory' using a dedicated set of address and data buses, and is connected to the 'data memory' using a different set of address and data buses.
- The Harvard architecture is used extensively in embedded systems, for example in digital signal processing (DSP) systems.
- Many microcontroller devices use a Harvard like architecture. Some types of microprocessor use a modified Harvard architecture.

Harvard Architecture

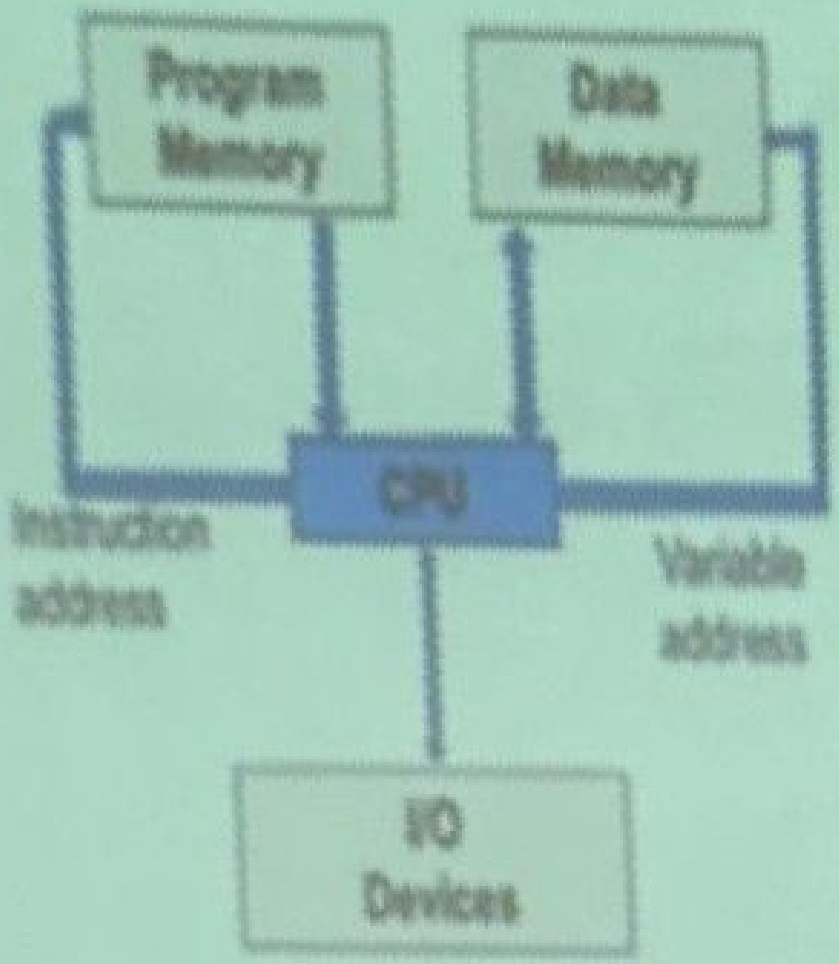




Harvard Architecture



Von Neumann Machine



Harvard Machine